Emerald Insight

International Journal of Clothing Science and Technology

Clothing similarity computation based on TLAC

Zheng Liu Jin Wang Qijin Chen Guodong Lu

Article information:

To cite this document:
Zheng Liu Jin Wang Qijin Chen Guodong Lu, (2012),"Clothing similarity computation based on TLAC", International Journal of Clothing Science and Technology, Vol. 24 Iss 4 pp. 273 - 286
Permanent link to this document:
http://dx.doi.org/10.1108/09556221211232856

Downloaded on: 30 January 2016, At: 07:42 (PT)
References: this document contains references to 15 other documents.
To copy this document: permissions@emeraldinsight.com
The fulltext of this document has been downloaded 216 times since 2012*

Users who downloaded this article also downloaded:

Qijin Chen, Jituo Li, Zheng Liu, Guodong Lu, Xinyu Bi, Bei Wang, (2013),"Measuring clothing image similarity with bundled features", International Journal of Clothing Science and Technology, Vol. 25 Iss 2 pp. 119-130 http://dx.doi.org/10.1108/09556221311298619

Asli Aksoy, Nursel Ozturk, Eric Sucky, (2012),"A decision support system for demand forecasting in the clothing industry", International Journal of Clothing Science and Technology, Vol. 24 Iss 4 pp. 221-236 http://dx.doi.org/10.1108/09556221211232829

Roqaya Sadek, (2012),"Effect of fabric softener on properties of a single jersey knitted fabric made of cotton and spandex yarn", International Journal of Clothing Science and Technology, Vol. 24 Iss 4 pp. 251-272 http://dx.doi.org/10.1108/09556221211232847

Access to this document was granted through an Emerald subscription provided by emerald-srm:101358 []

For Authors

If you would like to write for this, or any other Emerald publication, then please use our Emerald for Authors service information about how to choose which publication to write for and submission guidelines are available for all. Please visit www.emeraldinsight.com/authors for more information.

About Emerald www.emeraldinsight.com

Emerald is a global publisher linking research and practice to the benefit of society. The company manages a portfolio of more than 290 journals and over 2,350 books and book series volumes, as well as providing an extensive range of online products and additional customer resources and services.

Emerald is both COUNTER 4 and TRANSFER compliant. The organization is a partner of the Committee on Publication Ethics (COPE) and also works with Portico and the LOCKSS initiative for digital archive preservation.

*Related content and download information correct at time of download.

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

The current issue and full text archive of this journal is available at
www.emeraldinsight.com/0955-6222.htm

Clothing similarity computation based on TLAC

Zheng Liu
The State Key Laboratory of CAD & CG, Zhejiang University, Hangzhou, China and
Department of Clothing, Jiaxing University, Jiaxing, China, and
Jin Wang, Qijin Chen and Guodong Lu
The State Key Laboratory of CAD & CG, Zhejiang University, Hangzhou, China

Received 17 October 2011
Accepted 12 January 2012

Abstract

Purpose – To enhance the pleasure experience of clothes shopping online, finding satisfactory clothing and similar clothing recommendations to customers should be available and accurate. The purpose of this paper is to present a method for automatically computing the similarity between two apparels and giving an effective recommendation.

Design/methodology/approach – Based on a tabular layout of article characteristics the authors built a clothing information model to describe clothing. The clothing attributes are classified according to excavating features of the model. After the proposal of the computation algorithm for various attributes, an efficient similarity computation method is developed to obtain similar clothes with the given cloth. To prevent error and information omission during the computation, the analytic hierarchy process method and entropy method are adopted by the integrated weights as a control.

Findings – Clothing is a non-rigid product which has a lot of crossover and complicated attributes and features. This paper found a tabular layout of article characteristics can explain the clothing clearly. Through experiments the authors found the weight of attributes to have a great influence on similar results during the similarity computation.

Originality/value – This paper presents a new way to describe clothing information, and present the algorithm for attributes computation.

Keywords Clothing similarity, Tabular layouts of article characteristics, Clothing attributes, Clothing information model, Clothing industry, Information modelling

Paper type Research paper

1. Introduction

E-commerce is having a great impact on the way we do business. Clothing is one of the biggest growth markets for online shopping. Seeking satisfied clothes online is difficult for consumers since it needs both a professional sight and abundant energy. Clothing recommendation can effectively reduce the waste of time and energy when shopping online.

For clothing recommendation in most web sites, there are provided tools for searching clothing. Mainly of applied search technologies sort a cluster of clothing based key words such as category, feature, and brand, etc. However, the key words and the attributes definition is too simple/single and vague/ambiguous to get satisfying

This work was supported by the Project of Public Technology Research in Industry of Zhejiang Province (No.2010C31001), Research Fund for the Doctoral Program of Higher Education of China (No. 20100101110025), and the Natural Science Foundation of Ningbo City (No.2010A610117).

International Journal of Clothing Science and Technology
Vol. 24 No. 4, 2012
pp. 273-286
© Emerald Group Publishing Limited
0955-6222
DOI 10.1108/09556221211232856

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

search results. That technologies only can be used as a large-scale pre-filter, cannot provide consumers with clothing recommendation precisely. The better way is case-based reason (CBR) (Finnie and Sun, 2002; Slonim and Schneider, 2001; Liao *et al.*, 1998), which provide an interesting approach to integrate problem solving and learning in a variety of domains. This method is applied in a number of fields include software and engineering industry. Similarity is central to CBR because case retrieval uses similarity. As the same, the similarity computation is a core part of clothing searching and recommendation in clothing CBR.

In this paper, we provide a method for automatically compute the similarity between two apparels. Therefore, given an apparel image, the system would automatically search out the apparels with high similarity with the input one. Such a technology can be used and benefit the industry of on-line clothing shopping.

2. Related work

There exist a number of similarity computation methods, mostly of which are based on ontology, semantic and case-based reasoning (Li *et al.*, 2003; Cilibrasi and Vitanyi, 2007; Finnie and Sun, 2002; Batet *et al.*, 2011; Resnik, 1999). In ontology and semantic study, semantic similarity or semantic distance is achieved with computation of network distance models and information theoretic models usually. Finnie and Sun (2002) introduced similarity, similarity relations, similarity measures, similarity metrics and their relationships. They tried to clear this confusion by providing a unified framework for similarity. Resnik (1999) gave a way of similarity computation through measuring the degree of information sharing based on information theoretic models. For the case-based reasoning method, the essence is similarity computation. Due to uncertainty of objective environment and complexity of the subjective decision process, description of the problem is often incomprehensive and incomplete. About how to define the characteristics similarity of the case for the information, Slonim and Schneider (2001) presented a range attributes and fuzzy attributes similarity computations models in case-based reasoning, Liao *et al.* (1998) proposed a hybrid similarity measure with a mixture of crisp and fuzzy features for retrieval of similar cases.

Similarity computation is a multi-criterion decision-making problem. The weight of the element is a key index which influences the consequence greatly. The analytic hierarchy process (AHP) was developed by Saaty (1980) to structure complex multi-criteria decision-making in business. It has been applied to resolving kinds of decision-making problems. Zakarian and Kusiak (1999) and Amiri (2010) applied AHP in the determination of the importance measures for individual team members and criteria weights. However, when the descriptions of linguistic variable are vague and the verbal attitudes on evaluation process contain ambiguity and multiplicity of meaning, AHP is ineffective (Lee, 2010; Torfi *et al.*, 2010). Entropy is a concept of thermodynamics, used as measurement of disorder state for physical system. Shannon (1948) has led to a universal acceptance of the theory that information is statistical in nature. It is widely used in the information area for describing the disorder degree of a system. As one of the excellent tools in decision making, entropy has been widely applied in many fields.

Similarity computation is an import part to network semantic search, optimization for case-based algorithm, rigid material products as mechanical design and so on. However, it has not ever been used in soft products such as clothing. The material and the composition in clothing are totally different from the mechanical products.

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Clothing has plenty of comprehensive attributes, in both engineering and art domain. Each of people has different perception on the same clothing. The semantic similarity used in tradition way is difficult to explain the full clothing information clearly, and the algorithm in applied clothing searching cannot provide efficiency and effectiveness tools.

Nevertheless, the searching function should be equally applicable in the garment industry to support the fashion e-commerce activities. Aim at this point, we develop a method of clothing similarity computation based on the tabular layout of article characteristics (TLAC) and give a recommendation case with the method.

The remainder of this paper is organized as follows (Figure 1). In Section 3, the clothing model based on TLAC is built and explained, the kinds of the attributes are sorted. In Section 4 the framework and procedure of clothing similarity compute are introduced. In Section 5, the clothing attributes similarity computation algorithm is introduced. In Section 6, AHP and entropy methods are introduced for acquiring the subjective and objective weight of the clothing attributes and attribute modules. An application case of computation is given in Section 7. Finally, conclusion and suggestions for the future studies are clarified.

3. Clothing model expression based on TLAC

The TLAC is a way to describe the product's detail. It can be a base of product modeling. According to ISO13584, products characteristics are a set of information which includes such as geometry characteristic, function characteristic and manufacture characteristic of product, component and parts. TLAC can define and differentiate products crucial characteristics and or ordains express format of character value. It can be used to build the modules system that can support retrieval and variant design of parts effectively.

To express the clothing clearly and compute conveniently, we take TLAC to describe clothing in detail. In this paper, TLAC of clothing is a table with fixed format which records the characteristics. The whole clothing is constituted by several modules. Each module represents a specific part of clothing feature, such as style, color and fabric. Each module is structured by several clothing attributes that describing clothing character in detail. The definition of clothing attributes in TLAC is expressed by $Atr_cloth = \{Atr_content, Atr_option, Atr_catergory, Atr_value, Atr_constrain\}$. Table I shows the whole clothing information model in detail.

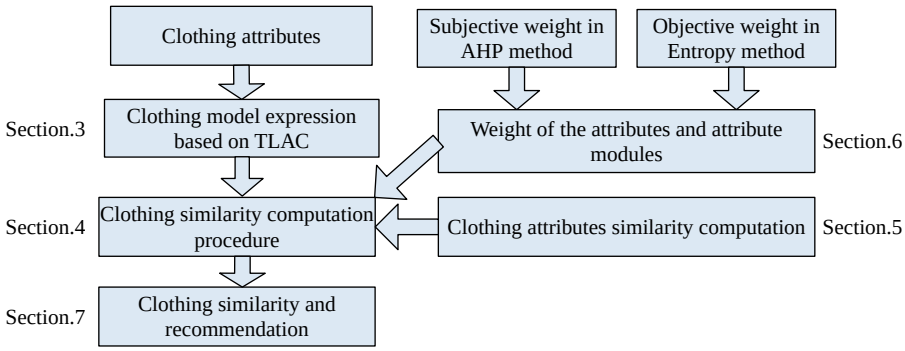

Figure 1. General scheme for clothing similarity computation

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Table I. Description of clothing in TLAC

Module_ID	Atr_content	Atr_option	Atr_catergory	Atr_value	Atr_constrain
Function	Season	Winter, spring, autumn, summer	Discontinuity	1-9	Dress occasion (business, business casual, leisure, lingerie)
	Gender	Male, neutral, female	Continuity	1-9	
	Top/bottom	Top, bottom, one-piece dress		1-9	
	Category	Vest, shirt, T-shirt, knitting, jacket, suit, overcoat; pant, skirt; one-piece shirt, one-piece pant	Discontinuity	1-9	
Style	Fittness	Tight, fit, loose	Continuity	1-9	Degree of style matching (strong, normal, weak)
	Silhouette	X, T, A, H, O	Discontinuity	1-9	
	Length	Short, normal, long	Explicit	0-1	
	Feature	Pattern, joint, cutting, double level, lace…	Discontinuity	1-9	
Color	Principal color	Ral code number(revision)	Continuity	100-923	Rule of color matching (color scheme, adjacent color, white-black, contrast color)
	Percentage	Number	Explicit	0-1	
	Number of minority color	Number	Explicit	0-1	
	minority color	RAL code number(revision)	Continuity	100-923	
Fabric	Principal material	Cotton, wool, silk, linen, viscose, polyester, spandex	Discontinuity	1-9	Reflection of the properties (principal, minority, blend)
	Thickness	Thin, normal, thick	Continuity	1-9	
	Minority material	Cotton, wool, silk, linen, viscose, polyester, spandex	Discontinuity	1-9	
	Weaving structure	Blending, interweave	Boolean	0 or 1	
–	–	–	–	–	–

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

As shown in Table I, "Atr_content", "Atr_option" and "Atr_value" are clothing detail description. "Atr_constrain" is not the same as the former, but about information matched with other attributes, is used to constrain the option selection, e.g. when "lingerie" is selected in the "Function" module, then "overcoat", "suit" and "jacket" are forbidden to be selected.

For describing attributes clearly and available in computation, types of "Atr_option" are divided into "Atr_catergory" as the following in detail: explicit type, can acquire the clothing attribute directly through inputting attribute values, such as length of clothing, percentage of fabric component; continuity type, which explains the relevance via professional knowledge, can describe attributes with value in accordance with a certain scale, such as dressing occasion (business to leisure), color (order in Ral color code); discontinuity type, denotes different detail features which is indirect relevance between two "Atr_option", can be converted to a symbol set for representing the clothing detail, such as style details, fabric characteristic and, etc.; Boolean type, express the clothing attributes only with two certain options and values them by 0 or 1, such as weaving structure, etc.

According to the "Atr_option" type category, the "Atr_value" is can be assigned as the following: explicit type attribute's primitive value is a linear range number, can be scaled within [0, 1]. Continuity type can be described by a range number which represent continuous degree of the attribute's feature. For evaluating degree conveniently, it can be scaled within [1, 9]. The feature of continuity type decides that we cannot find a continuous range number or character to represent its options. This type can only be flagged as a symbol. For mapping conveniently, we use the number of scale [1, 9] to represent it. Boolean type attribute value only has two choices 0 or 1.

In a word, TLAC can make clothing information clearly and available for calculability. Compared with other method, our approach can diminish the ambiguous and confusion information in a large degree.

4. Clothing similarity computation procedure

Generally, the main content of similarity computation is products information compute and weight coefficient. We describe the intrinsic information of clothing in TLAC, and compute in compatible algorithms. This paper presents to describe clothing attributes and weights in matrix or vector, and obtain the similarity from matrix operation progressively. Proposed similarity computation procedure is as below:

(1) We take a vectors: cloth $= \{a_1, \ldots, a_i, \ldots, a_m\}$ as the clothing information model, where m is the number of modules, each column $a_i = \{x_{i1}, \ldots, x_{ij}, \ldots, x_{in}\}$T represents a module, x_{ij} is the value of the jth attribute in ith module. For compute conveniently, the attribute value of clothing may be conversed to standardized one through $c_{ij} = x_{ij}/\sqrt{\sum_{j=1}^{J} x_{ij}^2}$, where c_{ij} is the standard value of the jth attribute in ith module, J is the number of apparels in the database. Then we get a normalized clothing matrix shown as:

$$\mathrm{C} = Clothing(c_{ij})_{n\times m} = \begin{pmatrix} c_{11} & \ldots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{n1} & \cdots & c_{nm} \end{pmatrix}$$

(2) For two clothing $\mathbf{C}_i$ and $\mathbf{C}_j$, their similarity can be computed by matching their TLACs which can be computed by matrix operations as would be detailed as below:

$$\lambda_{ij} = Sim(\mathbf{C}_i, \mathbf{C}_j) = Sim_M(\mathbf{C}_i, \mathbf{C}_j)\mathbf{W}_M = \begin{bmatrix} S_1 & S_2 & \cdots & S_m \end{bmatrix} \begin{bmatrix} w_1 & w_2 & \cdots & w_m \end{bmatrix}^T$$

where λ_{ij} represents the clothing similarity, Sim_M ($\mathbf{C}_i$, $\mathbf{C}_j$) denotes the similarity of attribute module between two clothing. For modules in TLAC, each of which has its certain contributions in identifying an individual clothing, which can be represented as a weight vector $\mathrm{W}_M = (w_{1,\ldots} w_{i,\ldots} w_m)^T$:

$$Sim_M(\mathbf{C}_i, \mathbf{C}_j) = \mathbf{W} Sim_P(\mathbf{C}_i, \mathbf{C}_j) = \begin{pmatrix} \omega_{11} & \ldots & \omega_{1n} \\ \vdots & \ddots & \vdots \\ \omega_{m1} & \cdots & \omega_{mn} \end{pmatrix} \oplus \begin{pmatrix} s_{11} & \ldots & s_{1m} \\ \vdots & \ddots & \vdots \\ s_{n1} & \cdots & s_{nm} \end{pmatrix} = \begin{bmatrix} S_1 & \cdots & S_m \end{bmatrix}$$

where $Sim_P(\mathbf{C}_i, \mathbf{C}_j)$ is the similarity of corresponding attributes pairs for $\mathbf{C}_i$ and $\mathbf{C}_j$, and $\mathbf{W} = (\omega_{ij})_{m \times n}$ represents the contributions of attributes in each modules:

$$Sim_P(\mathbf{C}_i, \mathbf{C}_j) = \begin{pmatrix} c_{11} & \ldots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{n1} & \cdots & c_{nm} \end{pmatrix}_i \otimes \begin{pmatrix} c_{11} & \ldots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{n1} & \cdots & c_{nm} \end{pmatrix}_j = \begin{pmatrix} s_{11} & \ldots & s_{1m} \\ \vdots & \ddots & \vdots \\ s_{n1} & \cdots & s_{nm} \end{pmatrix}$$

where the operator $\otimes$ means attributes similarity compute algorithm between two clothing, would be detailed explained in Section 5. The operator $\oplus$ represents multiply the similarity by weight. The definition shown as follows:

$$A \oplus B = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \oplus \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{pmatrix} = [\, a_{11} \cdot b_{11} + a_{12} \cdot b_{21} \quad a_{21} \cdot b_{12} + a_{22} \cdot b_{22} \,]$$

5. Clothing attributes similarity computation

From the description in Section 3, we sort the attributes to four types according their different features, and define the approach of value assignment. In Section 4, the attributes value is further normalized within [0, 1]. Based on this work, we present the attributes compute algorithms between two clothing, the operator $\otimes$ is as below:

(1) *Explicit and continuity type*. These attributes are linear in a range, can be deal with the nearest neighbor algorithm to get similarity as below:

$$Sim(c_i, c_j) = 1 - \frac{|c_i - c_j|}{\beta - \alpha}$$

where c_i and $c_j \in [\alpha, \beta]$, are the corresponding attributes in two clothing. α is the min value and β is the max value in the clothing attribute set:

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

(2) *Discontinuity type*. Value of this type just means a representation. We cannot calculate with the nearest neighbor. This paper takes the fuzzy mathematics algorithm to compute the similarity of this type, the formula shown as follows:

$$Sim(c_i, c_j) = \frac{c_i \cap c_j}{c_i \cup c_j} = \frac{c_i \cap c_j}{c_i + c_j - c_i \cap c_j}$$

(3) *Boolean type*. As attributes are represented in two dimensions' options, the similarity can be defined only with 1 and 0 in Boolean calculation, shown as follows:

$$Sim(c_i, c_j) = \begin{cases} 1, c_i = c_j \\ 0, c_i \neq c_j \end{cases}$$

In summary, if only using the nearest neighbor for the attributes similarity computation would lead to information missing and a poor quality result. Compared the traditional method, we practice a policy of divide and rule, adopt the nearest neighbor retrieval and fuzzy similarity retrieval method to compute the similarity, respectively, get a reasonable and easy result.

6. Weights in clothing similarity computation

Clothing similarity computation is a multi-criterion decision-making problem. The similarity weight is a key index which influences the consequence greatly. In this section, we take a series of weights to support the computation procedure. As mentioned in Section 2, the AHP takes the experts' experiences as well as deciders' preference and intention into account, brings a high reasonable order of attributes weight. However, the problem of arbitrariness exists inevitably when the evaluation process contain ambiguity and vague. To avoid the subjective effect, Jian *et al.* (1999) made an integrated approach to determine attribute weights in the multiple attribute decision-making problems, and took a sensitivity analysis through setting different percentage of subjective and objective weights. And entropy is a good way of evaluating the information objectively. Lin *et al.* (2010) applied the entropy weight-based decision-making theory to optimization of schemes for black-start in power systems. However, the representing the willing of experts and deciders is poor, the exported weight result may do not match the importance of actual fact, even be opposite.

In order to obtain an accurate weight, we determine $\mathbf{W}$ by integrating AHP method and entropy method, and $\mathbf{W}_M$ by AHP method, respectively. $\mathbf{W}$ is a linear combination of a subjective matrix $\mathbf{W}_s$ and objective matrix $\mathbf{W}_o$. Correspondingly, $\mathbf{W}_s$ is obtained by AHP method, $\mathbf{W}_o$ is obtained by entropy method. $\mathbf{W}_s$ is essentially required to make up the imperfection of $\mathbf{W}_o$. After plenty of experiments, we have found it is appropriate for setting α to 0.6:

$$\mathbf{W} = \alpha \mathbf{W}_s + (1 - \alpha)\mathbf{W}_o, \quad \alpha \in [0, 1]$$

6.1 Subjective weight based on AHP

The AHP method in this paper is applied to set $\mathbf{W}_M$ and $\mathbf{W}$. The steps shown are following:

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Step 1. Building a hierarchy structure, clothing attribute modules and attributes are converted to objectives elements in a hierarchical system as the follows:

$$U = \{u_1, u_2, \ldots, u_i, \ldots, u_n\}, \quad u_i \in U, \ i = 1, 2, \ldots, n$$

Set U represent clothing or attributes modules. Decomposed clothing or attributes modules and the hierarchy are constructed according to TLAC:

Step 2. In each hierarchical level, pair-wise comparisons of n attribute are made by using a scale 1-9. The value u_{ij} is assigned to represent the judgment concerning the relative importance of decision attribute u_i over u_j. Then, comparisons compose a pair-wise comparison matrix.

Step 3. In order to obtain the index of relative importance of clothing attributes, take the priority vector (or eigenvector) method to compute the attribute weight. From Perron-Frobenius theorem, the comparison matrix is calculated based on solving the equation:

$$AW = \lambda_{\max} Ws$$

where λ_{max} is the largest eigen-value, weight of compared clothing attribute is obtained from the component column of vector Ws.

6.2 Objective weight based on entropy method

We use entropy to measure the information variation of each item in the TLAC among the samples, and set the objective weights for the items one by one. For the items with high frequency variation, they are assigned high objective weights; otherwise they are assigned low weights. The entropy method is only applied to set W. Set each of clothing has m evaluating attributes indices, n denotes the number of evaluated clothing in set, and the weight of attribute is obtained shown as the following:

Step 1. Given an evaluating clothing set, a matrix with m attributes indices and n evaluated clothing for each, attribute modules is built. The matrix is translated to $R = (r_{ij})_{m \times n}$ with standardization. r_{ij} denotes the value in the ith index related to jth evaluated object, $r_{ij} \in [0,1]$. It is calculated by:

$$r_{ij} = \frac{\max(x_{ij}) - x_{ij}}{\max\limits_j (x_{ij}) - \min\limits_j (x_{ij})}$$

where x_{ij} denotes the origin matrix value in the ith index related to jth evaluated object:

Step 2. According to the definition of entropy, in an evaluating matrix with m evaluating attribute indices and n evaluated clothing, the entropy E_i of the ith evaluating index is defined as:

$$E_i = -\frac{1}{\ln \ n} \sum_{j=1}^{n} f_{ij} \ \ln \ f_{ij}, \quad i = 1, 2, \ldots, m$$

where $f_{ij} = r_{ij/} \sum_{j=1}^{n} r_{ij}$, $f_{ij} \ \ln \ f_{ij}$ would be zero if f_{ij} is equal to zero:

Step 3. The entropy weight $(\mathbf{W}_o)_i$ of the ith evaluating index can be calculated as:

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

$$\mathbf{W}_{\mathrm{o}} = \frac{1 - E_i}{m - \sum_{i=1}^{m} E_i}$$

where E_i is a nonnegative value, so $(\mathbf{W}_{\mathrm{o}})_i \in [0,1]$ and $\sum_i^m (\mathbf{W}_{\mathrm{o}})_i = 1$.

7. Case study: recommendation of clothing similarity computation

7.1 Clothing recommendation system based on TLAC

For evaluating the performance of our approach, we have developed an on-line clothing recommendation system. A number of clothing data were downloaded from the internet. Each of clothing has an image and detailed information such as material compositions and styles and sizes, etc. and this information is used to set corresponding TLAC tables. The system is composed with four parts: clothing data input, clothing attribute management, clothing search and client interface. The first three parts are achieved on the service. Figure 2 shows the interface for inputting the clothing data. In the figure, the right part shows the image of the inputting clothing, and the middle part is the corresponding TLAC table to be fulfilled. The clothing attribute management part provides tools to extend and edit the TLAC table, if necessary.

The clothing search part is the core part of the system, where our similarity computation algorithm is transplanted. On client interface, a user can choose either of following two modes to search clothing. One is by fulfilling a TLAC table; the other is by clicking an existing apparel which is mapped to a certain TLAC table in the database. Figure 3 shows the client interface.

The interface is composed with four parts as shown in Figure 3. The information of the picked apparel including the TLAC data and the image is detailed in a dialog as Part A shown. The data of TLAC is editable in the background. Regarding the picked apparel as sample clothing, when the green button in the dialog is pressed, the system would search the database for the apparels with high similarity to the sample apparel. Part B is the index of ranked result. The resulting apparels are displayed along a circle

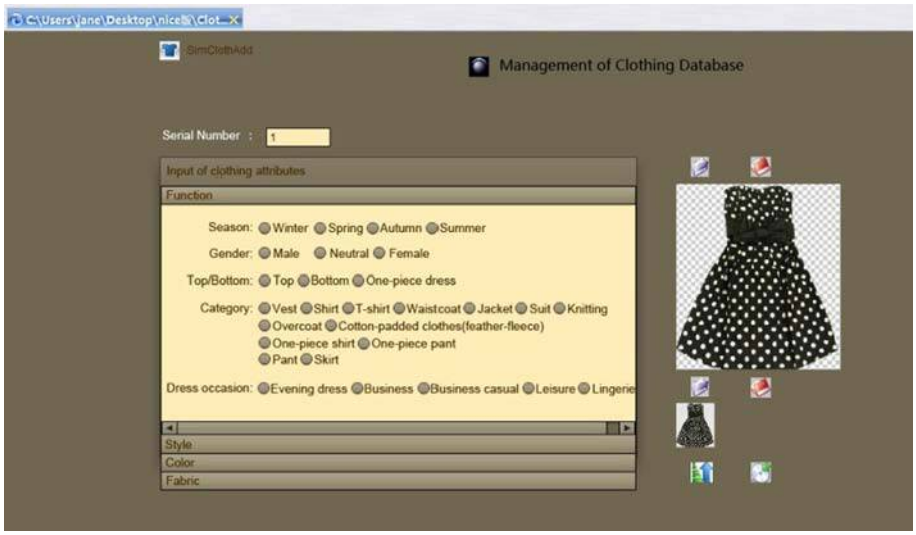

Figure 2.
Clothing data input

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

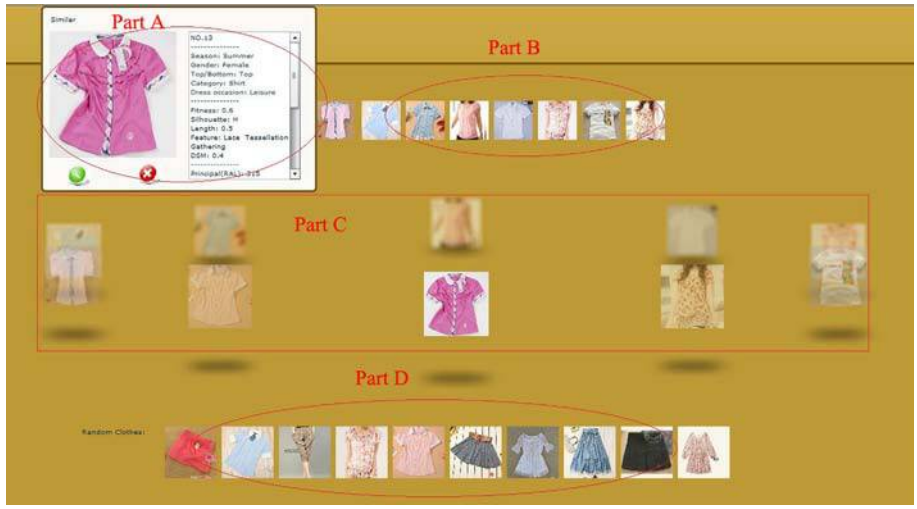

Figure 3.
Clothing display in client

as Part C shown in the middle row. The user can click each one to check the detailed information of the apparel to see whether it fits her/his requirements or not. Part D in the bottom row displays a number of clothing images in the database by random, among which the user select a favourite one by clicking the image.

We randomly selected out six clothing with their images from the database. The similarity of each pair of these clothing are computed and the most five similar clothing listed as shown in Figure 4. The first row is the seven target clothing, and the below five rows are the most similar clothing images and their similarity values. From the similar results, we can conclude that our algorithm performs well in clothing searching.

7.2 Comparison

In order to assess the effective of our approach's, some experiments and comparisons should be taken. However, it is difficult to compare our approach's performance and the other applied technologies because the clothing databases are different, and not normal. What is more, the ways of searching clothing is quite various. Actually, most of web site technologies search clothing by sorted based key words. Figure 5 is the interface of eBay women's clothing, which provides many key words or styles' options to users. Customers can obtain their favorites by progression choice. If the user demands are not clear yet, few favors can be done in the situation. Our approach support the recommendation of that favors an apparel on a model in a magazine/web through similarity computation as Figure 3, this is totally different.

In spite of the difficulty we also tried to prove the performance of our approach and evaluate the result compared with other ways. We took the 61 apparel as a sample, invited five fashion designers and five consumers to evaluate the similarity of between the objective clothing and the clothing database, and sorted the results by the similar value as right part of Table II. Then we acquire the similarity value through our approach as left part of Table II. The max degree of member is the final result of ten persons. From the results comparison, we can find that our approach can meet people's evaluation well especially in the top similar clothing.

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

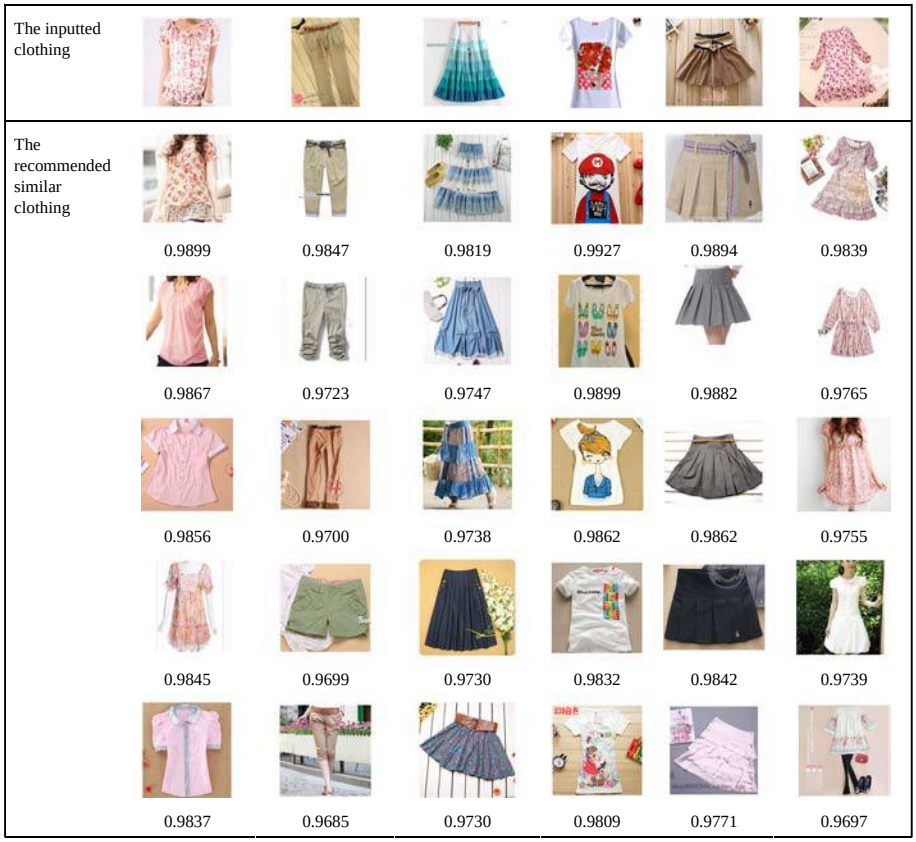

Figure 4.
Similarity calculation results of six different clothing styles

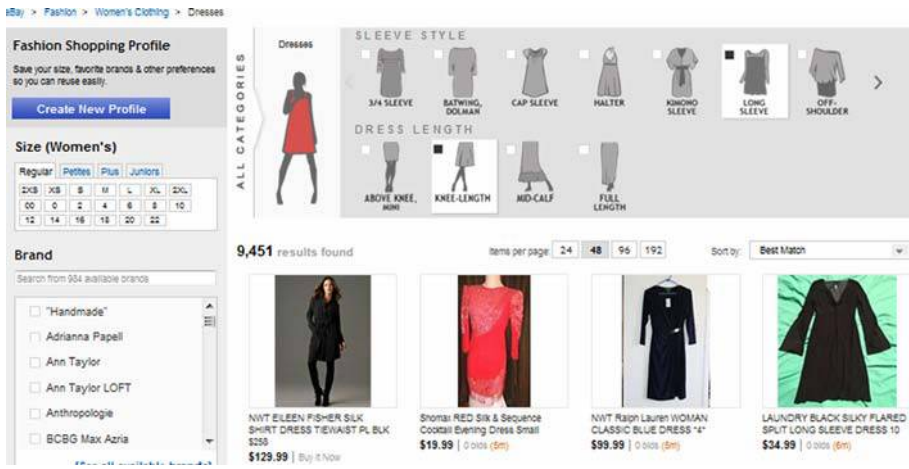

Figure 5.
The interface of eBay women's clothing

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Table II. Result comparison between the approach and ten persons

Order	Similar result by the approach: ID number	Similar result by the approach: Similarity value	Similarity result by ten persons: ID number										Max degree of member
1	27	0.9816	62	27	27	62	27	27	27	27	27	27	27
2	62	0.9837	27	62	34	27	62	35	62	62	62	62	62
3	34	0.9754	34	34	62	34	34	62	35	34	35	34	34
4	35	0.9750	35	35	35	29	23	34	34	35	34	35	35
5	23	0.9715	25	29	23	35	29	23	29	29	29	29	29
6	29	0.9695	33	23	29	23	35	29	23	23	23	23	23
7	30	0.9684	23	33	30	25	30	25	25	30	30	30	30
8	25	0.9682	30	25	25	33	33	30	33	25	25	25	25
9	33	0.9675	29	30	33	30	25	33	30	33	33	33	33

8. Conclusion

Considering that clothing online is a great prospect market and existing clothing search is not effective and ideal, we proposed a method of clothing similarity computation based on TLAC. The main advantages of this approach are following.

Based on describing attributes and attributes modules with the fixed form of TLAC, we explain the clothing information clearly and effectively, and make the clothing information calculated conveniently; with professional knowledge explanation, we present the algorithms of similarity computation for various types of attributes, make the customized computation according to attributes' categories; through integrating the weights of the AHP and entropy weight, we avoid the disadvantages of subjective and objective decision making, acquire the comprehensive weights.

As future research topics, efficiency and stability of our approach should be verified further in a large-scale database. Furthermore, customers are sometimes interested in one or few specific clothing features, therefore the computation of similarity weight should combine attributes weight and customers' preference information.

References

Amiri, M.P. (2010), "Project selection for oil-fields development by using the AHP and fuzzy TOPSIS methods", *Expert Systems with Applications*, Vol. 37 No. 9, pp. 6218-24.

Batet, M., Sanchez, D. and Valls, A. (2011), "An ontology-based measure to compute semantic similarity in biomedicine", *Journal of Biomedical Informatics*, Vol. 44 No. 1, pp. 118-25.

Cilibrasi, R.L. and Vitanyi, P.M.B. (2007), "The Google similarity distance", *IEEE Transactions on Knowledge and Data Engineering*, Vol. 19 No. 3, pp. 370-83.

Finnie, G. and Sun, Z.H. (2002), "Similarity and metrics in case-based reasoning", *International Journal of Intelligent Systems*, Vol. 17 No. 3, pp. 233-7.

Jian, M., Fan, Z.P. and Huang, L.H. (1999), "A subjective and objective integrated approach to determine attribute weights", *European Journal of Operation Research*, Vol. 112 No. 2, pp. 397-404.

Lee, S.H. (2010), "Using fuzzy AHP to develop intellectual capital evaluation model for assessing their performance contribution in a university", *Expert Systems with Applications*, Vol. 37 No. 7, pp. 4941-7.

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Li, Y.H., Bandar, Z.A. and Mclean, D. (2003), "An approach for measuring semantic similarity between words using multiple information sources", *IEEE Transactions on Knowledge and Data Engineering*, Vol. 15 No. 4, pp. 871-82.

Liao, T.W., Zhang, Z.M. and Mount, C. (1998), "Similarity measures for retrieval in case-based reasoning systems", *Applied Artificial Intelligence*, Vol. 12 No. 4, pp. 237-8.

Lin, Z.Z., Wen, F.S. and Huang, J.S. (2010), "Evaluation of black-start schemes employing entropy weight-based decision-making theory", *Journal of Energy Engineering*, Vol. 136 No. 2, pp. 42-9.

Resnik, P. (1999), "Semantic similarity in a taxonomy: an information-based measure and its application to problems of ambiguity in natural language", *Journal of Artificial Intelligence Research*, Vol. 11, pp. 95-130.

Saaty, T.L. (1980), *The Analytic Hierarchy Process: Planning, Priority Setting, Resources Allocation*, McGraw-Hill, London.

Shannon, C.E. (1948), "A mathematical theory of communication", *Bell System Technical Journal*, Vol. 27 No. 4, pp. 623-56.

Slonim, T.Y. and Schneider, M. (2001), "Design issues in fuzzy case-based reasoning", *Fuzzy Sets and Systems*, Vol. 117 No. 2, pp. 251-67.

Torfi, F., Farahani, R.Z. and Rezapour, S. (2010), "Fuzzy AHP to determine the relative weights of evaluation criteria and fuzzy TOPSIS to rank the alternatives", *Applied Soft Computing*, Vol. 10 No. 2, pp. 520-3.

Zakarian, A. and Kusiak, A. (1999), "Forming teams: an analytical approach", *IIE Transactions*, Vol. 31 No. 1, pp. 85-97.

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

Appendix

A sample of the similarity of between Nos 86 and 87 in the database is following:

- *Step 1: data normalization.* Through the equation $c_{ij} = x_{ij}/\sqrt{\sum_{j=1}^{J} x_{ij}^2}$, we convert the inputting TLAC value to standard one, e.g. the TLAC value of season options is: 1 for winter, 3 for spring and autumn and 5 for summer, so the standard value of summer is 0.7538, so is the other. The matrixes of standard value are built.
- *Step 2: the attributes' similarity computation.* According to Scetion 5, the corresponding attributes of two apparels is computed one by one, e.g. the attribute similarity of fitness is: $1 - (0.6 - 0.5)/1 = 0.9$, and the attribute of the fabric character is: $FC(86) \cap FC(87)/FC(86) \cup FC(87) = 1/1 = 1$, so is the other. Then the matrix of attributes' similarity between 86 and 87can be acquired:

$$\begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 1 \\ 0.814 & 0.9 & 1 & 1 & 0.635 \\ 1 & 0.899 & 1 & 1 & 1 \end{bmatrix}$$

- *Step 3: weights computation.* As mentioned in Section 6, the attributes module weight is obtained by AHP. A modules weight of this database is [0.657, 0.1911, 0.076, 0.076]. The attributes weight 1 and 2 are obtained by AHP and entropy, respectively. The results for this database are:

Attributes weights 1(AHP):

$$\begin{bmatrix} 0.5606 & 0.2124 & 0.1242 & 0.0392 & 0.0635 \\ 0.2084 & 0.1404 & 0.0940 & 0.2819 & 0.2753 \\ 0.4906 & 0.2337 & 0.0686 & 0.0686 & 0.1386 \\ 0.3864 & 0.2186 & 0.0550 & 0.0934 & 0.2467 \end{bmatrix}$$

Attributes weights 2(entropy):

$$\begin{bmatrix} 0.3349 & 0.4431 & 0.0179 & 0.0274 & 0.1764 \\ 0.1312 & 0.0894 & 0.0419 & 0.6724 & 0.0649 \\ 0.0579 & 0.0211 & 0.1092 & 0.7668 & 0.0446 \\ 0.1097 & 0.0223 & 0.3641 & 0.1397 & 0.3641 \end{bmatrix}$$

And the final attributes weights can be computed by $W = aW1 + (1 - a)\ W2$:

$$\begin{bmatrix} 0.4703 & 0.3047 & 0.0817 & 0.0344 & 0.1087 \\ 0.1775 & 0.1200 & 0.0732 & 0.4381 & 0.1912 \\ 0.3176 & 0.1487 & 0.0849 & 0.3479 & 0.1010 \\ 0.2757 & 0.1401 & 0.1786 & 0.1119 & 0.2937 \end{bmatrix}$$

- *Step 4.* Final similarity result for two apparels can be computed as Section 4, and the value is 0.9904323970862359.

Corresponding author
Jin Wang can be contacted at: dwjcom@zju.edu.cn

To purchase reprints of this article please e-mail: **reprints@emeraldinsight.com**
Or visit our web site for further details: **www.emeraldinsight.com/reprints**

Downloaded by Carleton University At 07:42 30 January 2016 (PT)

This article has been cited by:

1. Qijin Chen, Jituo Li, Zheng Liu, Guodong Lu, Xinyu Bi, Bei Wang. 2013. Measuring clothing image similarity with bundled features. *International Journal of Clothing Science and Technology* **25**:2, 119-130. [Abstract] [Full Text] [PDF]

Downloaded by Carleton University At 07:42 30 January 2016 (PT)